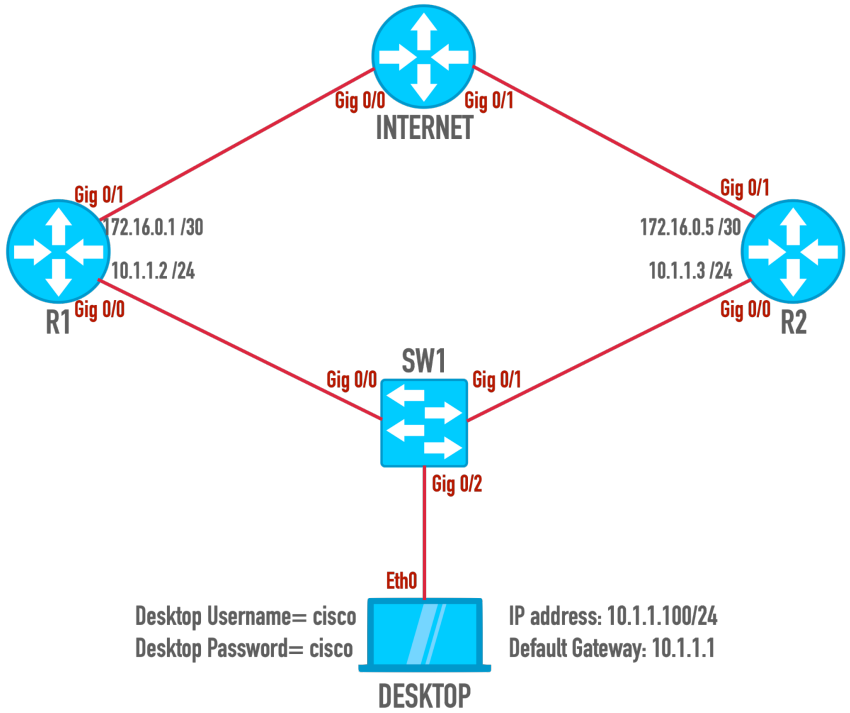
HSRP

Topology



Initial Configuration Commands

PC1:

```
sudo ifconfig eth0 10.1.1.100 netmask 255.255.255.0 up
sudo ip route default via 10.1.1.1
```

SW1:

```
enable
conf t
no ip domain-lookup
logging console
line con 0
logging synchronous
exec-timeout 0 0
hostname SW1
end
copy run star
```

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R1:

```
enable
conf t
no ip domain-lookup
logging console
line con 0
logging synchronous
exec-timeout 0 0
hostname R1
int gig 0/0
no shutdown
ip address 10.1.1.2 255.255.255.0
int gig 0/1
no shutdown
ip address 172.16.0.1 255.255.255.252
router ospf 1
network 10.1.1.0 0.0.0.255 area 0
network 172.16.0.0 0.0.0.3 area 0
end
copy run star
```

R2:

```
enable
conf t
no ip domain-lookup
logging console
line con 0
logging synchronous
exec-timeout 0 0
hostname R2
int gig 0/0
no shutdown
ip address 10.1.1.3 255.255.255.0
int gig 0/1
no shutdown
ip address 172.16.0.5 255.255.255.252
router ospf 1
network 10.1.1.0 0.0.0.255 area 0
network 172.16.0.0 0.0.0.3 area 0
end
copy run star
```

INTERNET:

```
enable
conf t
no ip domain-lookup
logging console
line con 0
logging synchronous
exec-timeout 0 0
hostname INTERNET
int gig 0/0
no shutdown
ip address 172.16.0.2 255.255.255.252
int gig 0/1
no shutdown
ip address 172.16.0.6 255.255.255.252
int lo 0
ip address 1.1.1.1 255.255.255.255
router ospf 1
network 172.16.0.0 0.0.0.3 area 0
network 1.1.1.1 0.0.0.0 area 0
exit
end
copy run star
```

Lab Tasks

- Configure R1 as the ACTIVE HSRP router for a virtual IP address of 10.1.1.1, by giving it a Priority of 110. Also, enable the Preempt feature.
- Configure R2 as the STANDBY HSRP router for the virtual IP address of 10.1.1.1. Also, enable the Preempt feature.
- Do a Trace Route from the PC to the "Internet" address of 1.1.1.1, and notice which gateway it's using to get out to the Internet.
- Start a continuous Ping to 1.1.1.1 on the DESKTOP.
- Shutdown interface Gig 0/0 on R1.
- Determine if any Ping packets were dropped after you shutdown the Gig 0/0 interface on R1. Hit “control C” to stop the continual pings 1.1.1.1.
- Do a Trace Route from the DESKTOP to the "Internet" address of 1.1.1.1, and notice which gateway it's using to get out to the Internet.
- Bring the Gig 0/0 interface on R1 back up.

Solution

Step 1: Configure R1 as the ACTIVE HSRP router for a virtual IP address of 10.1.1.1, by giving it a Priority of 110. Also, enable the Preempt feature.

```
R1>en
R1#conf t
R1(config)#int gig 0/0
R1(config-if)#standby 10 ip 10.1.1.1
R1(config-if)#standby 10 priority 110
R1(config-if)#standby 10 preempt
R1(config-if)#end
```

Step 2: Configure R2 as the STANDBY HSRP router for the virtual IP address of 10.1.1.1. Also, enable the Preempt feature.

```
R2>en
R2#conf t
R2(config)#int gig 0/0
R2(config-if)#standby 10 ip 10.1.1.1
R2(config-if)#standby 10 preempt
R2(config-if)#end
R2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Gi0/0       10   100 P Standby 10.1.1.2        local           10.1.1.1
(##R2 is the standby router)
```

Step 3: Do a Trace Route from the PC to the "Internet" address of 1.1.1.1, and notice which gateway it's using to get out to the Internet.

inserthostname-here login: **cisco**
Password: **cisco**

inserthostname-here:~$ **traceroute 1.1.1.1**
traceroute to 1.1.1.1 (1.1.1.1), 30 hops max, 46 byte packets
1 10.1.1.2 (10.1.1.2) 2.168 ms 2.310 ms 2.010 ms
2 172.16.0.2 (172.16.0.2) 2.646 ms 2.720 ms *
(##packets were sent to the INTERNET via R1)

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Step 4: Start a continuous Ping to 1.1.1.1 on the DESKTOP.

inserthostname-here:~$ **ping 1.1.1.1**
PING 1.1.1.1 (1.1.1.1): 56 data bytes
64 bytes from 1.1.1.1: seq=0 ttl=42 time=2.687 ms
64 bytes from 1.1.1.1: seq=1 ttl=42 time=4.859 ms
64 bytes from 1.1.1.1: seq=2 ttl=42 time=2.577 ms
64 bytes from 1.1.1.1: seq=3 ttl=42 time=2.476 ms
64 bytes from 1.1.1.1: seq=4 ttl=42 time=2.750 ms
64 bytes from 1.1.1.1: seq=5 ttl=42 time=2.697 ms
64 bytes from 1.1.1.1: seq=6 ttl=42 time=3.177 ms
64 bytes from 1.1.1.1: seq=7 ttl=42 time=2.721 ms
64 bytes from 1.1.1.1: seq=8 ttl=42 time=2.701 ms
64 bytes from 1.1.1.1: seq=9 ttl=42 time=2.665 ms
64 bytes from 1.1.1.1: seq=10 ttl=42 time=2.593 ms

Step 5: Shutdown interface Gig 0/0 on R1.

```
R1#conf t
R1(config)#int gig 0/0
R1(config-if)#shutdown
R1(config-if)#end
```

Step 6: Determine if any Ping packets were dropped after you shutdown the Gig 0/0 interface on R1. Hit "control C" to stop the continual pings.

inserthostname-here:~$ **ping 1.1.1.1**
PING 1.1.1.1 (1.1.1.1): 56 data bytes
64 bytes from 1.1.1.1: seq=0 ttl=42 time=2.687 ms
64 bytes from 1.1.1.1: seq=1 ttl=42 time=4.859 ms
64 bytes from 1.1.1.1: seq=2 ttl=42 time=2.577 ms
64 bytes from 1.1.1.1: seq=3 ttl=42 time=2.476 ms
64 bytes from 1.1.1.1: seq=4 ttl=42 time=2.750 ms
64 bytes from 1.1.1.1: seq=5 ttl=42 time=2.697 ms
64 bytes from 1.1.1.1: seq=6 ttl=42 time=3.177 ms

--- 1.1.1.1 ping statistics ---
51 packets transmitted, 17 packets received, 66% packet loss
round-trip min/avg/max = 2.831/3.314/5.362 ms

(##66% packet loss shows that we lost connection with R1 when we shutdown the Gig 0/0 interface)

Step 7: Do a Trace Route from the DESKTOP the "Internet" address of 1.1.1.1, and notice which gateway it's using to get out to the Internet.

```
inserthostname-here:~$ traceroute 1.1.1.1
traceroute to 1.1.1.1 (1.1.1.1), 30 hops max, 46 byte packets
 1  10.1.1.3 (10.1.1.3)  2.128 ms  2.869 ms  1.952 ms
 2  10.1.1.3 (10.1.1.3)  2.566 ms !H  2.128 ms !H
```

(##notice how packets are now being routed to R2)

Step 8: Bring the Gig 0/0 interface on R1 back up.

```
R1#conf t
R1(config)#int gig 0/0
R1(config-if)#no shutdown
R1(config-if)#end
```

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